



NSW Education Standards Authority

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Centre Number

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Student Number

2024 HIGHER SCHOOL CERTIFICATE EXAMINATION

Food Technology

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Write your Centre Number and Student Number at the top of this page

Total marks:
100

Section I – 20 marks (pages 2–6)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 50 marks (pages 9–18)

- Attempt Questions 21–26
- Allow about 1 hour and 25 minutes for this section

Section III – 15 marks (page 33)

- Attempt Question 27
- Allow about 30 minutes for this section

Section IV – 15 marks (page 34)

- Attempt Question 28
- Allow about 30 minutes for this section

Section I

20 marks

Attempt Questions 1–20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

- 1** Which sector of the Australian food industry has responsibility for the growth of seaweed?

 - A. Food retail
 - B. Agriculture and fisheries
 - C. Food service and catering
 - D. Food processing/manufacturing

- 2** Which of the following is a diet-related condition currently on the rise in Australia due to the over-consumption of foods high in kilojoules and fats?

 - A. Obesity
 - B. Anaemia
 - C. Osteoporosis
 - D. Type 1 diabetes

- 3** What is a me-too product?

 - A. A unique product
 - B. A completely new product
 - C. An extension of an existing product
 - D. A direct copy of an existing product

- 4 Which part of the marketing plan is associated with the advertising of a new food item?
- A. Price structure
 - B. Product life-cycle
 - C. Promotional program
 - D. Place and distribution
- 5 A student sells cupcakes at a school market day.
- Under which level of operation does this fall?
- A. Household
 - B. Multinational
 - C. Small business
 - D. Large company
- 6 A fast-food chain makes several changes to its menu earning it the Heart Foundation Tick.
- This results in consumers being provided with
- A. cheaper options.
 - B. healthier options.
 - C. more convenient foods.
 - D. special dietary products.
- 7 What step in the food product development process involves storage trials?
- A. Idea generation
 - B. Feasibility study
 - C. Product specifications
 - D. Testing product prototypes

- 8** A company that manufactures plant-based meat alternatives has decided to use newly available processing equipment.

What is probably the driver of this decision?

- A. Market concerns
 - B. Societal changes
 - C. Consumer demand
 - D. Technological developments
- 9** Food Standards Australia New Zealand (FSANZ) has made it mandatory for bread flour in Australia to be fortified with what additional ingredient?
- A. Iron
 - B. Calcium
 - C. Folic Acid
 - D. Vitamin C
- 10** Which food additive is used in table salt to prevent clumping, allowing it to flow?
- A. A humectant
 - B. An emulsifier
 - C. A bulking agent
 - D. An anti-caking agent
- 11** A bag of dried oats has been infested with insects.
- What is the most likely cause of this?
- A. Enzymatic activity
 - B. Environmental factors
 - C. Microbial contamination
 - D. Expiry of best-before date

- 12** A food company is being influenced by rising interest rates.

Which macro-environment is likely to be most affected?

- A. Economic
- B. Ecological
- C. Legislative
- D. Technological

- 13** Which of the following is produced using ecologically-sustainable production methods?

- A. Skim milk
- B. Caged eggs
- C. Organic apples
- D. Freeze-dried berries

- 14** An Australian energy drink company is targeting a new drink at children by using bright colours and celebrity endorsements on social media.

Which of the following is responsible for overseeing this company's ethical compliance?

- A. World Health Organisation
- B. The Advertising Standards Bureau
- C. Australian Food and Grocery Council
- D. Food Standards Australia New Zealand

- 15** Quality control measures are used to ensure that raw materials used in the manufacturing process are

- A. in season.
- B. value for money.
- C. conveniently sourced.
- D. meeting specifications.

- 16** Which of the following lists two health issues related to the over-consumption of some vitamin supplements?
- A. Kidney failure and liver damage
 - B. Constipation and haemorrhoids
 - C. Coeliac disease and diarrhoea
 - D. Dental caries and bad breath
- 17** Which of the following is the primary purpose of centrifuging as a processing technique?
- A. Enhancing flavour
 - B. Increasing temperature
 - C. Separating components
 - D. Reducing cooking time
- 18** Which active non-nutrient is beneficial for an individual recovering from diverticulitis?
- A. Probiotics
 - B. Dietary fibre
 - C. Phyto-chemicals
 - D. Omega-3 fatty acids
- 19** Which legislation ensures compliance of the safe operating of machinery in the NSW food industry?
- A. *Food Act 2003*
 - B. *Food Regulation Act 2015*
 - C. *Work Health and Safety Act 2011*
 - D. *Competition and Consumer Act 2010*
- 20** What temperature and time are required to produce shelf stable, long-life milk?
- A. 110°C for 3 seconds
 - B. 110°C for 32 seconds
 - C. 140°C for 3 seconds
 - D. 140°C for 32 seconds

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Centre Number

Food Technology

Sections II, III and IV

Answer Booklet

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Student Number

Instructions

- Write your Centre Number and Student Number at the top of this page.
- Use this answer booklet for Sections II, III and IV.
- Follow the instructions inside this booklet as to where you should write your answers.

Please turn over

Section II

50 marks

Attempt Questions 21–26

Allow about 1 hour and 25 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Question 21 (11 marks)

Name an organisation in the Australian food industry that you have studied.

Name of organisation:

- (a) Identify TWO career opportunities within the organisation. 2

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- (b) Describe the level of operation and mechanisation within the organisation. 4
Provide a relevant example.

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Question 21 continues on page 11

Question 21 (continued)

- (c) Explain the impact that this organisation has on the Australian economy.

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End of Question 21

Please turn over

Question 22 (8 marks)

- (a) Describe the potential risks and benefits of ONE emerging technology for the food service and catering sector.

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Question 22 continues on page 13

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Question 22 (continued)

- (b) Explain how organic farming addresses the environmental concerns associated with food production.

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End of Question 22

Please turn over

Question 23 (12 marks)

- (a) Why is food colouring used as an additive in the production of lollies? **2**

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- (b) A multinational company produces biscuits 24 hours a day, 7 days a week. **2**

Identify TWO characteristics of the equipment required for this production.

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- (c) Describe how food is preserved using fermentation. Provide a relevant example. **4**

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Question 23 continues on page 15

Question 23 (continued)

- (d) Explain the significance of quality management in ensuring the safety of frozen foods. Provide a relevant example.

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End of Question 23

Please turn over

Question 24 (9 marks)

- (a) Compare TWO types of food product development.

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- (b) Why is it important to test product prototypes when creating a new food product?

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Question 25 (6 marks)

Discuss the role that supplements can have in the diet of a professional athlete.
Provide relevant examples.

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Please turn over

Question 26 (4 marks)

The diagram shows the ingredients used in the popular drink *bubble milk tea*.

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Bubble Milk Tea

Explain the potential health implications for an adolescent who consumes 500 mL of this beverage daily.

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Section III — Answer Question 27 on pages 19–25

27

Question Number

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– 23 –

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If you require more space to answer Section III, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Section III, tick here.

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Section IV — Answer Question 28 on pages 26–32

28

Question Number

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If you require more space to answer Section IV, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Section IV, tick here.

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Food Technology

Section III

15 marks

Attempt Question 27

Allow about 30 minutes for this section

Answer the question on pages 19–25 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Question 27 (15 marks)

A food company has introduced a new tropical fruit and avocado drink that is lactose-free and enriched with active non-nutrients.

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|-----|--|----------|
| (a) | Identify TWO drivers that could have influenced the development of this product. | 2 |
| (b) | Explain how the food company's micro-environment could affect the success of this drink. | 5 |
| (c) | Explain the health benefits of this drink for an individual. | 8 |

Please turn over

Section IV

15 marks

Attempt Question 28

Allow about 30 minutes for this section

Answer the question on pages 26–32 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Your answer will be assessed on how well you:

- demonstrate knowledge and understanding relevant to the question
 - apply course concepts to food technology issues
 - communicate ideas and information using appropriate terminology and relevant examples
 - present a logical and cohesive response
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Question 28 (15 marks)

Explain how food manufacturers use preservation processes and current developments in packaging to respond to consumer demands.

End of paper